Yeshi Dolma

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Research Interest

Computational AI for Public Health Research and Translation: Applied Machine Learning Scientist with 9 years of experience designing and deploying large-scale AI systems in production. Currently focused on advanced predictive modeling, interpretable AI, and causal inference to identify mechanisms driving public health patterns, women's health outcomes and translate complex health data into policy-relevant insights and evidence for public health systems.

Interests: Predictive and counterfactual modeling; resource optimization; spatiotemporal systems; behavioral health analytics; global health; women's health and reproductive health; cancer research; bias-aware causal inference; and scalable AI for low-resource public health systems.

Education

Indian Institute of Technology, Kanpur (IIT Kanpur)

Integrated Dual Degree (B.Tech. & M. Tech.), Computer Science & Engineering | 2011 - 2016

GPA: 8.3/10; Equivalent: 3.5/4

Master's Thesis: ***Zero-Shot Learning of Image Categories using Gaussian Processes***
Supervisor: Dr. Vinay P. Namboodiri

- **Core Contribution:** Developed a novel probabilistic architecture for **Zero-Shot Learning (ZSL)**, enabling robust classification for image categories where no training examples exist.
- **Methodology:** Pioneered a hybrid approach using Gaussian Process modeling to bridge the gap between visual features and semantic attributes, significantly enhancing the model's ability to generalize to novel categories.
- **Results:** Demonstrated superior performance in predictability and interpretability compared to traditional ZSL baselines on benchmark datasets.
- **Publication**: Peer-reviewed and published in the **ACCV 2016** (tier-1 conference) and **Springer** (2017). Presented at the **Grace Hopper** Conference India (2017).

Academic Minor: English Literature | GPA: 10/10; Equivalent: 4/4

Publications

- **Dolma, Y.**, & Namboodiri, V. P. (2016). Using Gaussian Processes to Improve Zero-Shot Learning with Relative Attributes. *Proceedings of the Asian Conference on Computer Vision (ACCV), Taipei, Taiwan.*
- **Dolma, Y.**, Kalani, R., Agrawal, A., & Basu, S. (2021). Improving Bounce Rate Prediction for Rare Queries by Leveraging Landing Page Signals. *Proceedings of the Web Conference Workshop on Transforming Big Data into Actionable Knowledge (BiDAW), Ljubljana, Slovenia.*

Independent Projects in Public Health & Safety

Alternative Response Systems Optimisation: Public Health Strategy & Optimization
Seattle, WA | 2026

- **Crisis Triage Framework:** Developed an incident-first classification system to isolate the 6.4% of 911 CAD events involving behavioral health crises. Identified a critical coverage gap where only 1.19% of eligible incidents received a health-centered (CARE) response, providing the empirical basis for expanding non-police interventions.
- **Structural Determinants Analytics:** Quantified crisis burden using Racial & Social Equity (RSE) indices; found high-vulnerability neighborhoods experience **3× higher crisis rates**, driven by socioeconomic factors ($\rho \approx 0.44$).

- **Structural Gap Diagnosis:** Revealed that **80.4%** of CARE-eligible crises default to police response due to geographic availability rather than eligibility or triage issue. Benchmarked "Success Hubs" against "Service Gaps" neighbourhood to identify high-marginal-benefit sectors for targeted team placement.
- **Preventative Integration:** Mapped neighborhoods with structurally limited early-crisis identification. Proposed non-police care strategies to position alternative response as a preventative public health intervention, targeting triggers before escalation and downstream harm.
- **Resource Allocation Framework:** Presented a place-based deployment strategy to CARE leadership team that balances need and operational efficiency through residential and activity-driven placement. Identified field-level access mechanisms like SPD-CARE on-view coordination and non-police/social-worker presence as critical complements to geographic expansion for improving accessibility.
- Abstract submitted to the **American Public Health Association (APHA) 2026** annual meeting.

COVID-19 Misinformation Dynamics and Policy-Linked Public Health Impacts | 2025

- **Causal Analysis Pipeline:** Engineered a pipeline to analyze Google Trends as proxy for misinformation engagement (e.g., vaccine denial, institutional skepticism) and correlated composite indices with COVID-19 trajectories in Washington.
- Developed workflows for signal smoothing, normalization, and policy-event annotation to systematically contextualize and isolate the drivers of misinformation surges.
- **Mechanism Identification:** Revealed that misinformation engagement was primarily driven by **policy and communication vacuums** (e.g., CDC mask guidance shifts, vaccine rollout gaps), and guidance reversals while remaining largely decoupled from disease severity; **institutional skepticism** persisted due to cumulative credibility erosion.
- **Policy-Relevant Insights:** Research indicated that effective interventions require targeting **institutional communication** and **behavioral strategies** over traditional models focused solely on addressing disease-based information gaps.
- Extending the study with NLP (**topic modeling, sentiment analysis**) to trace narrative evolution in misinformation headlines and examine variation across states with **differing social and demographic contexts**.

Early Detection of High-Risk User Intent through Search Session Analysis | 2017

- **Public Safety Gap Analysis:** Identified critical failures in search engine responses where indirect or escalating crisis intent bypassed standard helpline triggers, often returning unsafe results for high-risk users.
- **Temporal Intent Modeling:** Modeled multi-query search sessions to capture escalation patterns in depressive and high-risk intent, shifting the detection paradigm from isolated keywords to behavioral sequences.
- **Early Intervention Impact:** Demonstrated that session-level modeling identifies crisis signals significantly earlier than query-level triggers, establishing a scalable framework for digital mental health interventions.

Smart ICU Monitoring Prototype (GE Healthcare Hackathon – 2nd Prize) | 2017

- **Clinical Vision System:** Prototyped a computer vision–based safety system for intensive care units, leveraging depth-camera signals to monitor patient posture, movement, and distress indicators.
- **Real-Time Signal Processing:** Engineered Python-based algorithms for automated fall and distress detection, integrating real-time alerting logic to demonstrate technical feasibility in clinical settings.
- **Translational Innovation:** Awarded **Second Prize** at the GE Healthcare Hackathon for the system's potential to improve digital health monitoring and patient outcomes.

Work and Research Experience

Microsoft Corporation, **Applied Scientist | August 2016 – December 2025**

Signals Team, Microsoft Advertising- Machine Learning and Prediction Systems

- **Large-Scale Predictive Modeling:** Designed, trained, and productionized advanced machine-learning systems (deep neural nets, transformer-based models, gradient-boosted trees, fast rank, extreme classifications) to predict user engagement signals (CTR, bounce-back, audience response) at global scale across high-throughput advertising platforms.
- **End-to-End ML Systems Ownership:** Architected full ML lifecycles spanning data ingestion, feature generation, model training, offline evaluation, online serving, and continuous monitoring, operating under strict latency, reliability, and scalability constraints.
- **Privacy-Preserving ML & Regulatory Compliance:** Designed modeling workflows compliant with GDPR and global privacy standards, integrating data anonymization and access controls into long-running production systems.
- **Causal-Aware Experimentation & Evaluation:** Led large-scale experimentation using randomized A/B tests and observational analyses to measure incremental lift, explicitly addressing confounding, position bias, selection effects, and feedback loops inherent in ad-ranking systems.
- **Representation Learning & Feature Design:** Developed semantic, contextual, and behavioral representations (embeddings and structured features) capturing user intent and interaction dynamics, enabling robust generalization under sparse and shifting data regimes.
- **Responsible AI & Model Governance:** Built drift, bias, and performance-degradation detection frameworks with diagnostic dashboards, ensuring sustained model reliability and alignment with Microsoft's Responsible AI principles in live production environments.
- **Research Contributions:** Published **applied ML research** The Web Conference Workshop BiDAW '21 (**tier-1** venue) and led translation of research prototypes into globally deployed productionized systems.

Selected Research Projects at Microsoft

Modeling Bounce Back Rate Prediction for Ranking Algorithm

- **Led modeling** and global deployment of bounce-back rate prediction, a core engagement signal powering Bing ranking and bid optimization across multilingual markets.
- Introduced **contextual and semantic feature representations,** improving performance on sparse and long-tail queries; mentored an intern through model development and evaluation.
- **First-author publication** at The Web Conference (WWW) Workshop 2021, demonstrating significant AUC gains on rare queries by mitigating **sparse-data challenges** through novel feature design.
- **Optimized** end-to-end production pipelines, reducing model inference latency by 4–10% via targeted feature pruning and system-level tuning.

Extreme Classification Methods for Complex Audience Segmentation

- Applied **Parabel extreme classification** to large-scale multi-label prediction, addressing high-dimensional label spaces and scalability constraints.
- **Unified hundreds of logistic regression models** into a single extreme-classification framework, improving generalization while simplifying system complexity.
- Achieved **20–25% performance gains** on sparse audience-segment prediction, enabling more efficient and robust downstream targeting.

Query Embedding Representation Learning for Sparse and Diverse Data

- **Developed neural embedding models** to represent user queries in a shared low-dimensional semantic space, supporting language- and market-agnostic understanding.
- Improved Click-Through-Rate prediction and robustness for **previously unseen or low-history queries** by transferring signal via semantic similarity to known patterns.
- **Trained terabyte-scale embeddings** using distributed ML frameworks (Scope, Aether), optimizing for efficient training and low-latency online serving.
- Demonstrated how **representation learning mitigates sparsity and improves generalization** in large-scale retrieval and ranking system.

Achievements and Awards

- **Q4 Greatness Award** – One Microsoft (2021): Recognized for cross-team collaboration and global product delivery.
- **Q4 Greatness Award – Engineering Excellence**, Microsoft Advertising (2020).
- Recognition of **Commitment & Contribution Award**, Microsoft Advertising (2017).
- **Top Design Idea**, Microsoft Code.Fun.Do (2016).
- **Women Emerging in Finance Event**, Goldman Sachs (2015).
- **AISSCE State Topper (2010):** served as **state valedictorian.** Ranked first across **all schools** in the state with **95.2%.**
- **NTSE Scholar** (2008): National-level merit scholarship by **National Council of Educational Research and Training**, India.
- **Nationwide Interactive Math Olympiad** (2008).

Selected Course work

Public Health & Epidemiology | *Johns Hopkins University - Online*

- Infectious Disease Transmission Models for Decision-Makers
- Essential Epidemiologic Tools for Public Health Practice
- Data and Health Indicators in Public Health Practice
- Foundations of Health Equity Research
- Design and Conduct of Clinical Trials
- Introduction to the GBD | ***IHME - Online***

Computer Science & Engineering | *Indian Institute of Technology, Kanpur*

- Machine Learning for Computer Vision
- Computer Vision and Image Processing
- Data Structures and Algorithms
- Design and Analysis of Algorithms
- Introduction to Computer Graphics
- Compiler Design
- Theory of Computation
- Algorithmic Game Theory
- Technical Communication
- Introduction to Logic
- Principles of Programming Languages
- Principles of Database Systems

- Advanced Computer Networks
- Software Architecture
- Computer Organization
- Linear Algebra
- Discrete Mathematics
- Probability and Statistics

Teaching Assistant | *Indian Institute of Technology, Kanpur*

- Machine Learning for Computer Vision (Spring 2016)
- Computer Science and Engineering Communication Skills (Fall 2015)